

Assignment 2

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Course : CSE 422

Section : 12

1) (a) cheat and pass are conditionally independent given study when

$$P(C \cap P | S) = P(C | S) \cdot P(P | S)$$

$$P(C | S) = \frac{P(C \cap S)}{P(S)} = \frac{0.25 + 0.02}{(0.25 + 0.15 + 0.02 + 0.03)} = 0.6$$

$$P(P | S) = \frac{P(P \cap S)}{P(S)} = \frac{(0.25 + 0.15)}{(0.25 + 0.15 + 0.02 + 0.03)} = 0.89$$

$$P(C | S) \cdot P(P | S) = 0.53$$

Again,

$$P(C \cap P | S) = \frac{P(C \cap P \cap S)}{P(S)} = \frac{0.25}{(0.25 + 0.15 + 0.02 + 0.03)} = 0.55$$

$$P(C \cap P | S) \neq P(C | S) \cdot P(P | S)$$

$\therefore$ So cheat and pass are not conditionally independent

$$(b) P(P \cup C) = P(P) + P(C) - P(P \cap C)$$

$$P(P) = (0.25 + 0.15 + 0.10 + 0.13) = 0.63$$

$$P(C) = (0.25 + 0.02 + 0.10 + 0.22) = 0.59$$

$$P(P \cap C) = (0.25 + 0.10) = 0.35$$

$$\therefore P(P \cup C) = (0.63 + 0.59 - 0.35) = 0.87 \quad (\text{Ans})$$

$$2) (a) P(\neg C) = (0.12 + 0.04 + 0.10 + 0.07) = 0.33$$

$$(b) P(\neg C | \neg R \cap \neg C) = \frac{P(\neg C \cap \neg R \cap \neg C)}{P(\neg R \cap \neg C)} = \frac{0.07}{(0.04 + 0.07)} = 0.63$$

$$(c) P(\neg R | \neg C) = \frac{P(\neg R \cap \neg C)}{P(\neg C)} = \frac{0.03 + 0.07}{(0.26 + 0.10 + 0.03 + 0.07)} = 0.21$$

$$(d) P(\neg R \cup C) = P(\neg R) + P(C) - P(\neg R \cap C)$$

$$P(\neg R) = (0.06 + 0.04 + 0.03 + 0.07) = 0.2$$

$$P(C) = (0.32 + 0.06 + 0.12 + 0.04) = 0.54$$

$$P(\neg R \cap C) = (0.06 + 0.04) = 0.1$$

$$P(\neg R \cup C) = (0.2 + 0.54 - 0.1) = 0.64 \quad (\text{Ans})$$

$$3)(a) P(F|L) = \frac{P(F \cap L)}{P(L)} = \frac{0.15}{(0.24 + 0.15 + 0.15)} = 0.28$$

$$(b) P(R|C) = \frac{P(R \cap C)}{P(C)} = \frac{0.1}{(0.24 + 0.1)} = 0.29$$

(c) Here, cricket and football have no relation. So they have 0 probability. $P(F \cap C) = 0$

(d) The probability of being Right handed or left-handed is 1. Because they have separate two columns having no overlapping relationship.

$$(e) P(F|R) = \frac{P(F \cap R)}{P(R)} = \frac{0.1}{0.1 + 0.1 + 0.26} = 0.21$$

$$P(F) = (0.15 + 0.1) = 0.25$$

$P(F|R) \neq P(F)$ [Football is dependant on Right Handed]

$$4) P(a \cap b \cap c) = \frac{1}{8} = 0.125$$

$$P(a \cap \neg b \cap c) = \frac{1}{8} = 0.125$$

$$P(\neg a \cap \neg b \cap c) = \frac{2}{8} = 0.25$$

$$P(a \cap \neg b \cap \neg c) = \frac{1}{8} = 0.125$$

$$P(a \cap b \cap \neg c) = \frac{3}{8} = 0.375$$

Joint distribution table

	b		$\neg b$	
	c	$\neg c$	c	$\neg c$
a	0.125	0.375	0.125	0.125
$\neg a$	0	0	0.25	0

$$P(\neg a | \neg b) = \frac{P(\neg a \cap \neg b)}{P(\neg b)}$$

$$= \frac{0.25}{(0.125 + 0.125 + 0.25)}$$

$$= 0.5$$

5) (a) D^L = Detect lie

D^T = Detect truth

L = Telling lie

$$P(L) = 0.02$$

$$P(\neg L) = 0.98$$

$$P(D^L | L) = 0.96$$

$$P(D^T | L) = (1 - 0.96) = 0.04$$

$$P(D^T | \neg L) = 0.95$$

$$P(D^L | \neg L) = (1 - 0.95) = 0.05$$

$$P(L | D^L) = \frac{P(D^L | L) \cdot P(L)}{P(D^L)} = \frac{0.96 \times 0.02}{0.02 \times 0.96 + 0.98 \times 0.05}$$

$$P(\neg L | D^L) = \frac{P(D^L | \neg L) \cdot P(\neg L)}{P(D^L)} = \frac{0.05 \times 0.98}{0.02 \times 0.96 + 0.98 \times 0.05}$$

Crossing out the denominators we get

$$P(L | D^L) = (0.96 \times 0.02) = 0.0192$$

$$P(\neg L | D^L) = (0.05 \times 0.98) = 0.049$$

$$\therefore P(\neg L | D^L) > P(L | D^L)$$

$\therefore$ The person is more likely to be not a liar

$$\begin{aligned}
 (b) P(D^+) &= P(D^+|L) \cdot P(L) + P(D^+|7L) \cdot P(7L) \\
 &= (0.96 \times 0.02) + (0.05 \times 0.98) \\
 &= 0.06
 \end{aligned}$$

$$\begin{aligned}
 P(7L|D^+) &= \frac{P(D^+|7L) P(7L)}{P(D^+)} \\
 &= \frac{0.05 \times 0.98}{0.06} \\
 &= 0.81
 \end{aligned}$$

$$\begin{aligned}
 6) (a) P(Y) &= \frac{5}{8} \\
 P(N) &= \frac{3}{8}
 \end{aligned}$$

$$\begin{aligned}
 P(Y|S, M, N, W) &= P(S, M, N, W|Y) \cdot P(Y) \\
 &= P(S|Y) \cdot P(M|Y) \cdot P(N|Y) \cdot P(W|Y) \cdot P(Y) \\
 &= \frac{2}{5} \times \frac{3}{5} \times \frac{4}{5} \times \frac{3}{5} \times \frac{5}{8} \\
 &= 0.07
 \end{aligned}$$

$$\begin{aligned}
 P(S|Y) &= \frac{2}{5} \\
 P(M|Y) &= \frac{3}{5} \\
 P(N|Y) &= \frac{4}{5} \\
 P(W|Y) &= \frac{3}{5} \\
 P(Y) &= \frac{5}{8}
 \end{aligned}$$

$$\begin{aligned}
 P(N_0 | S, M, N, W) &= P(S, M, N, W | N_0) \cdot P(N_0) \\
 &= P(S | N_0) \cdot P(M | N_0) \cdot P(N | N_0) \cdot P(W | N_0) \cdot P(N_0) \\
 &= \frac{3}{3} \times \frac{2}{3} \times 0 \times \frac{1}{3} \times \frac{3}{8} \\
 &= 0
 \end{aligned}$$

$$\therefore P(Y | S, M, N, W) > P(N_0 | S, M, N, W)$$

$\therefore$ Player is going to play tennis

$$\begin{aligned}
 (b) \quad P(Y | O, H) &= P(O, H | Y) \cdot P(Y) \\
 &= P(O | Y) \cdot P(H | Y) \cdot P(Y) \\
 &= \frac{2}{5} \times 0 \times \frac{5}{8} \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 P(N_0 | O, H) &= P(O, H | N_0) \cdot P(N_0) \\
 &= P(O | N_0) \cdot P(H | N_0) \cdot P(N_0) \\
 &= 0 \times \frac{1}{3} \times \frac{3}{8} \\
 &= 0
 \end{aligned}$$

Since both has 0 probability. We cannot infer the probability.

7) Given,

$$m=3$$

$$LR=0.01$$

$$b=1$$

$$y' = mx + b$$

For each value of x_i , we find y'

x	y	y'
2	14	7
4	29	13
6	42	19

$$y'_1 = (3 \times 2) + 1 = 7$$

$$y'_2 = (4 \times 3) + 1 = 13$$

$$y'_3 = (6 \times 3) + 1 = 19$$

$$MSE = \frac{1}{n} \sum (y_i - \hat{y}_i)^2$$

$$= \frac{1}{3} \{ (14-7)^2 + (29-13)^2 + (42-19)^2 \}$$

$$= 278$$

$$\frac{\partial L}{\partial m} = -\frac{2}{n} \sum x_i (y_i - \hat{y}_i)$$

$$= -\frac{2}{3} \{ 2(14-7) + 4(29-13) + 6(42-19) \}$$

$$= -144$$

$$\frac{\partial L}{\partial b} = -\frac{2}{n} \sum (y_i - \hat{y}_i)$$

$$= -\frac{2}{3} \times \{(14-7) + (29-13) + (42-19)\}$$

$$= -30.67$$

$$\text{new } m = m - \left(LR \times \frac{\partial L}{\partial m} \right)$$

$$= 3 - (0.01 \times -144)$$

$$= 4.44$$

$$\text{new } b = b - \left(LR \times \frac{\partial L}{\partial b} \right)$$

$$= 1 - (0.01 \times -30.67)$$

$$= 1.3067$$

10) (a) Given, $LR = 0.05$

$$m = 2$$

$$b = 1$$

$$y' = mx + b$$

$$y'_1 = 2 \times 1 + 1 = 3$$

$$y'_2 = 2 \times 3 + 1 = 7$$

$$y'_3 = 2 \times 5 + 1 = 11$$

x	y	y'
1	5	3
3	12	7
5	15	11

$$MSE = \frac{1}{n} \sum (y_i - \hat{y}_i)^2$$

$$= \frac{1}{3} \times \{ (5-3)^2 + (12-7)^2 + (15-11)^2 \}$$

$$= 15$$

$$(b) \frac{\partial L}{\partial m} = -\frac{2}{n} \sum n_i (y_i - \hat{y}_i)$$

$$= -\frac{2}{3} \{ 1(5-3) + 3(12-7) + 5(15-11) \}$$

$$= -24.67$$

$$\frac{\partial L}{\partial b} = -\frac{2}{n} \sum (y_i - \hat{y}_i)$$

$$= -\frac{2}{3} \{ (5-3) + (12-7) + (15-11) \}$$

$$= -7.33$$

$$new_m = 2 - (0.05 \times -24.67) = 3.2335$$

$$new_b = 1 - (0.05 \times -7.33) = 1.3665$$

$$\text{new } y_1'' = (3.2335 \times 1) + 1.3665 = 9.6$$

$$y_2'' = (3.2335 \times 3) + 1.3665 = 11.067$$

$$y_3'' = (3.2335 \times 5) + 1.3665 = 17.534$$

$$C = \frac{1}{3} \{ (5 - 9.6)^2 + (12 - 11.067)^2 + (15 - 17.534)^2 \}$$

$$= 9.48$$

$$\frac{dL}{dm} = -\frac{2}{3} \times \{ 0.4 + 0.9337 + (-2.54) \}$$

$$= 0.801$$

$$\therefore m_{\text{new}} = 3.2335 - (0.05 \times 6.314) = 2.918$$

$$\therefore b_{\text{new}} = 1.3665 - (0.05 \times 0.801) = 1.326 \quad (\text{Ans})$$

(c) $(y_i - y')$ would be more suitable. Gradient descent optimizes the values of m and b . If a loss function is taken which calculates the loss that can be neutralized. So if we take a positive loss function, we can accurately predict how close

we are to the values of m and b . $(y_i - y)^3$ might give negative values. So $|y_i - y|$ is better.

$$\begin{aligned} 11) (a) z &= w_1 x_1 + w_2 x_2 + b \\ &= 0.5 \times 1 + (-0.25 \times 2) + 0 \end{aligned}$$

$$\sigma(z) = \frac{1}{1+e^{-z}} = \frac{1}{1+e^{-0}} = 0.5$$

$$\therefore y' = 0.5$$

Given,

$$x_1 = 1, LR = 0.1$$

$$x_2 = 2$$

$$y = 0, b = 0$$

$$w_1 = 0.5, w_2 = -0.25$$

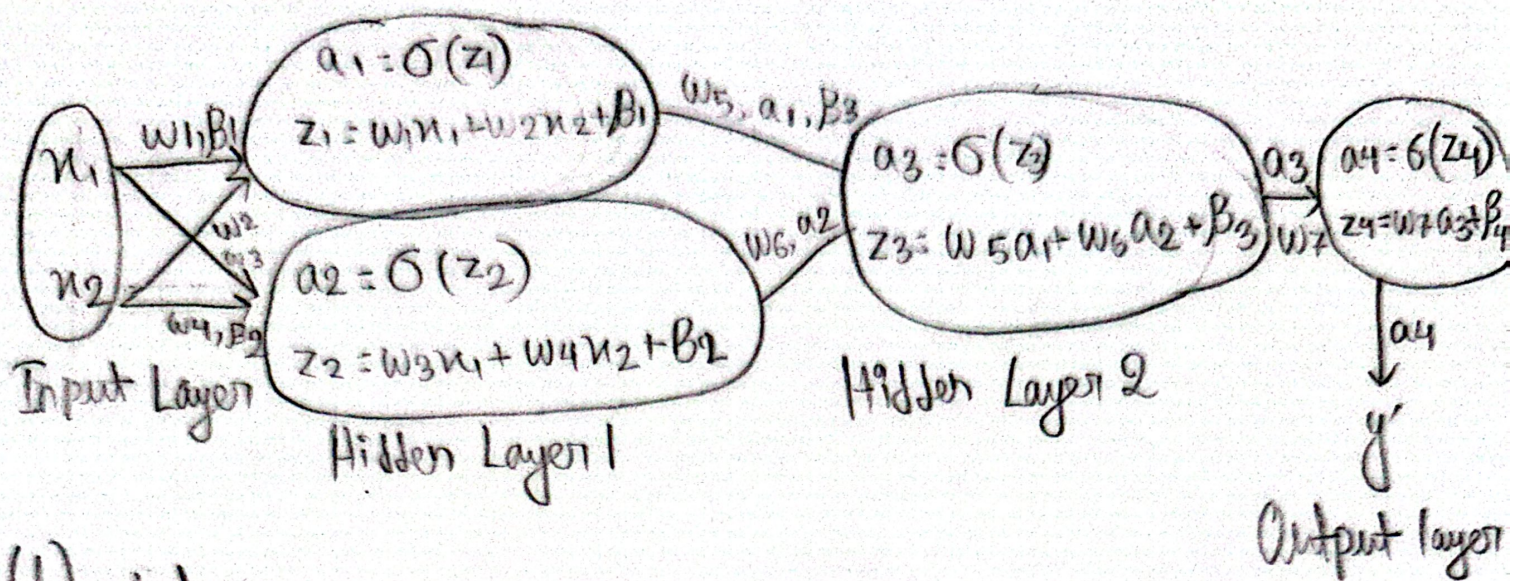
$$(b) \frac{\partial L}{\partial w} = x (y' - y)$$

$$\begin{aligned} \therefore \frac{\partial L}{\partial w} &= 1 \times (0.5 - 0) \\ &= 0.5 \end{aligned}$$

$$\therefore w_{\text{new}} = 0.5 - (0.1 \times 0.5) = 0.45$$

(c) Model can take larger steps towards global minimum and slows down when gets more closer to global min. through decaying learning rate.

12)(a)



(b) Let, $\beta_3 = 0$

Given, $a_1 = 0.5, a_2 = 0.5$

$w_5 = 0.1, w_6 = 0.5$

Now,

$$z_3 = w_5 a_1 + w_6 a_2 + \beta_3$$

$$= (0.1 \times 0.5) + (0.5 \times 0.5)$$

$$= 0.3$$

$$\therefore a_3 = \frac{1}{1 + e^{-z}} = \frac{1}{1 + e^{-0.3}} = 0.57$$

(c) We know each neuron has 1 activation output. And in the given stem, hidden layer 1 has 2 neurons, that means it has 2 activation outputs. On the other hand, hidden layer 2 has only 1 neuron. So it has only 1 activation output.